

OSSP TASK 1

```
lenovo@Manya:~$ pwd
/home/lenovo
lenovo@Manya:~$ cd Desktop
-bash: cd: Desktop: No such file or directory
lenovo@Manya:~$ cd ~
lenovo@Manya:~$ mkdir OSSP
mkdir: OSSP: File exists
lenovo@Manya:~$ cd OSSP
lenovo@Manya:~/OSSP$ pwd
/home/lenovo/OSSP
lenovo@Manya:~/OSSP$ mkdri src include onj bin docs tests scripts
Command 'mkdri' not found, did you mean:
  command 'mkdir' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
lenovo@Manya:~/OSSP$ mkdir src include obj bin docs tests scripts
mkdir: src: File exists
mkdir: include: File exists
mkdir: obj: File exists
mkdir: bin: File exists
mkdir: docs: File exists
mkdir: tests: File exists
lenovo@Manya:~/OSSP$ ls
assests  bin  docs  include  obj  scripts  scripts  src  tests
lenovo@Manya:~/OSSP$ cs src
Command 'cs' not found, but can be installed with:
sudo apt install csound
lenovo@Manya:~/OSSP$ cd src
lenovo@Manya:~/OSSP/src$ touc main.c shell.c parser.c executor.c builtin.c
Command 'touc' not found, did you mean:
  command 'touch' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'touch' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'touch' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'touch' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
lenovo@Manya:~/OSSP/src$ touch main.c shell.c parser.c executor.c builtin.c
lenovo@Manya:~/OSSP/src$ cd..
cd..: command not found
lenovo@Manya:~/OSSP/src$ cd include
```

```

lenovo@Manya:~/OSSP/src$ cd include
-bash: cd: include: No such file or directory
lenovo@Manya:~/OSSP/src$ ls
builtin.c executor.c main.c parser.c shell.c
lenovo@Manya:~/OSSP/src$ cd include
-bash: cd: include: No such file or directory
lenovo@Manya:~/OSSP/src$ lenovo@Manya:~/OSSP$
-bash: lenovo@Manya:~/OSSP$: No such file or directory
lenovo@Manya:~/OSSP/src$ cd ..
lenovo@Manya:~/OSSP$ cd include
lenovo@Manya:~/OSSP/include$ touch shell.h parser.h executor.h builtin.h
lenovo@Manya:~/OSSP/include$ ls
builtin.h executor.h parser.h shell.h
lenovo@Manya:~/OSSP/include$ cd .
-bash: cd: too many arguments
lenovo@Manya:~/OSSP/include$ cd ..
lenovo@Manya:~/OSSP$ cd docs
lenovo@Manya:~/OSSP/docs$
lenovo@Manya:~/OSSP/docs$ touch design.md report.pdf
lenovo@Manya:~/OSSP/docs$ cd ..
lenovo@Manya:~/OSSP$ cd tests
lenovo@Manya:~/OSSP/tests$ touch test_parser.c test_executor.c
lenovo@Manya:~/OSSP/tests$ cd ..
lenovo@Manya:~/OSSP$ cd scripts
lenovo@Manya:~/OSSP/scripts$ touch run.sh
lenovo@Manya:~/OSSP/scripts$ cd ..
lenovo@Manya:~/OSSP$ cd assets
-bash: cd: assets: No such file or directory
lenovo@Manya:~/OSSP$ mkdri src include onj bin docs tests scripts assets
Command 'mkdri' not found, did you mean:
  command 'mkdir' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'mkdir' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
lenovo@Manya:~/OSSP$ mkdir src include obj bin docs tests scripts assets
mkdir: src: File exists
mkdir: include: File exists
mkdir: obj: File exists
mkdir: bin: File exists
mkdir: docs: File exists

```

```

lenovo@Manya:~/OSSP/assets$ touch architecture.png
lenovo@Manya:~/OSSP/assets$
lenovo@Manya:~/OSSP/assets$
lenovo@Manya:~/OSSP/assets$ cd ..
lenovo@Manya:~/OSSP$ touch README.md LICENSE Makefile.gitignore
lenovo@Manya:~/OSSP$ ls -a
. . . LICENSE Makefile.gitignore README.md assests assets bin docs include obj scipts scripts src tests
lenovo@Manya:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/lenovo/OSSP/.git/
lenovo@Manya:~/OSSP$ ls -a
. . . .git LICENSE Makefile.gitignore README.md assests assets bin docs include obj scipts scripts src tests
lenovo@Manya:~/OSSP$ .git
Command '.git' not found, did you mean:
  command 'dgit' from deb dgit (15.5)
  command 'git' from deb git (1:2.53.0-1ubuntu1)
  command 'jgit' from deb jgit-cli (6.7.0-2build1)
  command 'qgit' from deb qgit (2.13-1)
Try: sudo apt install <deb name>
lenovo@Manya:~/OSSP$ sudo apt instaall trr
[sudo: authenticate] Password:
[sudo: authentication failed, try again.
[sudo: authenticate] Password:
Error: Invalid operation instaall
lenovo@Manya:~/OSSP$ sudo apt install tree
Installing:
  tree
Summary:

```

```

Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 25
Download size: 53.5 kB
Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 2s (26.2 kB/s)
Selecting previously unselected package tree.
(Reading database ... 35901 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
lenovo@Manya:~/OSSP$ tree
.
├── LICENSE
├── Makefile.gitignore
├── README.md
├── assests
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
└── scripts

```

OSSP TASK 2

```

Setting up cpp (4:13.2.0-5ubuntu1) ...
Setting up gcc (4:15.2.0-5ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
lenovo@Manya:~$ Enter command (or exit): date
-bash: syntax error near unexpected token '('
lenovo@Manya:~$ Enter command (or exit): date
-bash: syntax error near unexpected token '('
lenovo@Manya:~$ ^C
lenovo@Manya:~$ ^C
lenovo@Manya:~$ lenovo@Manya:~$
lenovo@Manya:~$: command not found
lenovo@Manya:~$ cd ProcessTask
lenovo@Manya:~/ProcessTask$ ls
task2.c
lenovo@Manya:~/ProcessTask$ Enter command (or exit):
-bash: syntax error near unexpected token '('
lenovo@Manya:~/ProcessTask$ date
Mon Aug  3 13:24:13 UTC 2026
lenovo@Manya:~/ProcessTask$ mkdir ProcessTask
cd ProcessTask

nano task2.c

gcc task2.c -o task2

./task2

```

```

Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [5840 B]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp amd64 4:15.2.0-5ubuntu1 [22.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcc1-0 amd64 16-20260322-1ubuntu1 [51.2 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgomp1 amd64 16-20260322-1ubuntu1 [162 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 libitm1 amd64 16-20260322-1ubuntu1 [30.0 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 libasan8 amd64 16-20260322-1ubuntu1 [3184 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblsan0 amd64 16-20260322-1ubuntu1 [1394 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 libtsan2 amd64 16-20260322-1ubuntu1 [2849 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 libubsan1 amd64 16-20260322-1ubuntu1 [1240 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 libhwasec0 amd64 16-20260322-1ubuntu1 [1731 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 libquadmath0 amd64 16-20260322-1ubuntu1 [155 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgcc-15-dev amd64 15.2.0-16ubuntu1 [2866 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [25.4 MB]
Get:19 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15 amd64 15.2.0-16ubuntu1 [539 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [1208 B]
Get:21 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc amd64 4:15.2.0-5ubuntu1 [5010 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-dev-bin amd64 2.43-2ubuntu2.3 [23.3 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-libc-dev amd64 7.0.0-28.28 [1619 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/main amd64 rpcsvc-proto amd64 1.4.3-1build1 [68.3 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dev amd64 2.43-2ubuntu2.3 [2295 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute/main amd64 manpages-dev all 6.17-1 [2183 kB]
Fetched 59.5 MB in 32s (1890 kB/s)
Selecting previously unselected package gcc-15-base:amd64.
(Reading database ... 35908 files and directories currently installed.)
Preparing to unpack .../00-gcc-15-base_15.2.0-16ubuntu1_amd64.deb ...
Unpacking gcc-15-base:amd64 (15.2.0-16ubuntu1) ...
Selecting previously unselected package libisl23:amd64.
Preparing to unpack .../01-libisl23_0.27-1build1_amd64.deb ...
Unpacking libisl23:amd64 (0.27-1build1) ...
Selecting previously unselected package libmpc3:amd64.
Preparing to unpack .../02-libmpc3_1.3.1-3_amd64.deb ...
Unpacking libmpc3:amd64 (1.3.1-3) ...
Selecting previously unselected package cpp-15-x86-64-linux-gnu.
Preparing to unpack .../03-cpp-15-x86-64-linux-gnu_15.2.0-16ubuntu1_amd64.deb ...
Unpacking cpp-15-x86-64-linux-gnu (15.2.0-16ubuntu1) ...
Selecting previously unselected package cpp-15.
Preparing to unpack .../04-cpp-15_15.2.0-16ubuntu1_amd64.deb ...
Unpacking cpp-15 (15.2.0-16ubuntu1) ...
Selecting previously unselected package cpp-x86-64-linux-gnu.
Preparing to unpack .../05-cpp-x86-64-linux-gnu_4%3a15.2.0-5ubuntu1_amd64.deb ...
Unpacking cpp-x86-64-linux-gnu (4:15.2.0-5ubuntu1) ...
Setting up cpp (4:15.2.0-5ubuntu1) ...
Setting up gcc (4:15.2.0-5ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
lenovo@Manya:~$ Enter command (or exit): date
-bash: syntax error near unexpected token '('
lenovo@Manya:~$ Enter command (or exit): date
-bash: syntax error near unexpected token '('
lenovo@Manya:~$ ^C
lenovo@Manya:~$ ^C
lenovo@Manya:~$ lenovo@Manya:~$
lenovo@Manya:~$: command not found
lenovo@Manya:~$ cd ProcessTask
lenovo@Manya:~/ProcessTask$ ls
task2.c
lenovo@Manya:~/ProcessTask$ Enter command (or exit):
-bash: syntax error near unexpected token '('
lenovo@Manya:~/ProcessTask$ date
Mon Aug 3 13:24:13 UTC 2026
lenovo@Manya:~/ProcessTask$ mkdir ProcessTask
cd ProcessTask

nano task2.c

gcc task2.c -o task2

./task2

```

GNU nano 8.7.1

task2.c

```
    if(pid < 0)
    {
        printf("Fork Failed\n");
    }
    else if(pid == 0)
    {
        execlp(command, command, NULL);

        printf("Invalid Command\n");
        exit(1);
    }
    else
    {
        wait(NULL);
    }
}

return 0;
}
```

Write to File:<L>; } } return 0;}}

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^Q Discard buffer	

```
lenovo@Manya:~$ gcc task2.c -o task2
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
lenovo@Manya:~$ sudo apt update
sudo apt install gcc
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
gcc

Installing dependencies:
cpp          libc-dev-bin  libmpc3
cpp-15       libc6-dev     libquadmath0
cpp-15-x86-64-linux-gnu libgcc1-0     libtsan2
cpp-x86-64-linux-gnu  libgcc-15-dev libubsan1
gcc-15        libgomp1      linux-libc-dev
gcc-15-base   libhwasan0    manpages-dev
gcc-15-x86-64-linux-gnu libisl23      rpcsvc-proto
gcc-x86-64-linux-gnu  libitm1
libasan8       liblsan0

Suggested packages:
cpp-doc      make      flex      gcc-15-multilib  glibc-doc
gcc-15-locales autoconf  bison     gcc-15-doc
cpp-15-doc   automake  gdb       gdb-x86-64-linux-gnu
gcc-multilib libtool   gcc-doc   libc-devtools

Summary:
Upgrading: 0, Installing: 26, Removing: 0, Not Upgrading: 30
Download size: 59.5 MB
Space needed: 200 MB / 1024 GB available

Continue? [Y/n]
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-base amd64 15.2.0-16ubuntu1 [39.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libisl23 amd64 0.27-1build1 [691 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmpc3 amd64 1.3.1-3 [54.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [12.9 MB]
```